

1. We prove that $n! > 3n$ for every natural number n such that $n \geq 4$ using the principle of Mathematical Induction.

Let the given statement be $P(n)$

$$P(n) : n! > 3n$$

Base case:

$$P(4) : 4! = 24 > 3 \times 4 = 12$$

Therefore $P(n)$ is true for $n=4$.

Let $P(n)$ is true for $n=m$.

$$P(m) : m! > 3m \quad - \quad (1)$$

Next we prove that $P(m+1)$ is true ~~when~~

$$\text{Since } m! > 3m$$

$$(m+1)m! > 3m(m+1)$$

$$(m+1)m! > 3(m+1)$$

$$(m+1)! > 3(m+1)$$

Therefore $P(m+1)$ is true whenever $P(m)$ is true.

2. Alice plays first. As the rules of the game says she has to remove atleast one coin from any of the two piles. Clearly choice of pile does not matter. So without loss of generality assume that she chose the first pile. Now she can remove any number of coins say, r such that $1 \leq r \leq n$.

Case 1: ($r = n$)

If she removes all the coins from one pile then Bob removes $(n-1)$ coins from the second pile leaving Alice no choice but to remove the last coin. Therefore, Bob definitely wins if Alice removes all the coins from one pile.

Case 2: ($r = n-1$)

If Alice removes $n-1$ coins from one pile. Which means that out of those two piles one contains exactly one whereas the other contains exactly n coins. Then Bob removes all the coins from the pile which contains n coins. So, now also Alice has to pick the last coin left.

Case 3: ($1 \leq r < n-1$)

If she removes r coins from the first pile then Bob removes r coins from the second pile. Now we have $(n-r)$ coins in both piles. Since the number of coins are decreasing, they will reach to Case 1 or Case 2. and then Bob will win.

3. For every $n \in \mathbb{N}$ and every tournament with n players, there exists a win-sequence $x_1, \dots, x_n$ that includes each of the n players in the tournament exactly once.

If we remove a player from a tournament with at least two players then we get a tournament with $n-1$ players.

Using previous problem a sequence of distinct players $x_1, \dots, x_r$ a win-sequence if either $r=1$ for every when

$$2 \leq i \leq r \quad \text{when} \quad n \geq 2$$

Initially the heap sizes are n and $2(k+1)$ and $2(k+1) \geq 4$ players shall split

$$2(k+1) \text{ int } 2(k) \text{ and } 2.$$

4. A tournament consists of n players and all possible pairwise matches between them. Denote the score vector of the tournament as $s = (s_1, s_2, s_3, \dots, s_n)$ and assume without loss of generality $s_1 > s_2 > s_3 > \dots > s_n$. It is also given that $2 \leq k \leq n$ for any,

$$s_1 + s_2 + s_3 + s_4 + \dots + s_n \geq C(k, 2) \text{ where}$$

$$C(k, 2) = \frac{k(k+1)}{2}$$

By induction we see that one of the two players win so if we remove two players the no. of players be $(n-1)$

And we clearly say that the win is player is $x_1, x_2, x_3, \dots, x_n$ because by given information that the no. of player is n and win like $x_1, x_2, x_3, \dots, x_n$ and we know one of two players can win it means if we remove two players then no. of players $(n-1)$ Then win $x_1, x_2, x_3, \dots, x_{n-2}$ Because one of two

Players can win not exactly win but there is no choice that every player will win.